2. Evaluate the following integrals using properties of integration :

(i) $\int_{-5}^{5} x\cos\left(\frac{e^x-1}{e^x+1}\right)dx$

(ii) $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (x^5 + x\cos x + \tan^3 x + 1)\,dx$

(iii) $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \sin^2 x\ dx$

(iv) $\int_{0}^{2\pi} x\log\left(\frac{3+\cos x}{3-\cos x}\right)dx$

(v) $\int_{0}^{2\pi} \sin^4 x\cos^3 x\,dx$

(vi) $\int_{0}^{1} |5x-3|\ dx$

(vii) $\int_{0}^{\sin^2 x} \sin^{-1}\sqrt{t}\ dt + \int_{0}^{\cos^2 x} \cos^{-1}\sqrt{t}\ dt$

(viii) $\int_{0}^{1} \frac{\log(1+x)}{1+x^2}\ dx$

(ix) $\int_{0}^{\pi} \frac{x\sin x}{1+\sin x}dx$

(x) $\int_{\frac{\pi}{8}}^{\frac{3\pi}{8}} \frac{1}{1+\sqrt{\tan x}}dx$

(xi) $\int_{0}^{\pi} x\left[\sin^2(\sin x) + \cos^2(\cos x)\right]dx$

9.4 Bernoulli's Formula

The evaluation of an indefinite integral of the form $\int u(x)v(x)dx$ becomes very simple, when u is a polynomial function of x (that is, $u(x) = a_0x^n + a_1x^{n-1} + \cdots + a_n$) and $v(x)$ can be easily integrated successively. It is accomplished by a formula called **Bernoulli's formula**. This formula is actually an extension of the formula of integration by parts. To derive the formula, we use the following notation:

$$u^{(1)} = \frac{du}{dx},\quad u^{(2)} = \frac{du^{(1)}}{dx},\quad u^{(3)} = \frac{du^{(2)}}{dx},\ \cdots\cdots$$

$$v_{(1)} = \int v dx,\ v_{(2)} = \int v_{(1)}dx,\ v_{(3)} = \int v_{(2)}dx,\cdots\cdots$$

Then, we have

$$dv_{(1)} = vdx,\ dv_{(2)} = v_{(1)}dx,\ dv_{(3)} = v_{(2)}dx,\ \cdots\cdots$$

Now, by integration by parts, we get

$$\int uvdx = \int udv_{(1)} = uv_{(1)} - \int v_{(1)}du = uv_{(1)} - \int v_{(1)}\frac{du}{dx}dx$$

$$= uv_{(1)} - \int u^{(1)}dv_{(2)}$$

$$= uv_{(1)} - \left(u^{(1)}v_{(2)} - \int v_{(2)}du^{(1)}\right)$$

$$= uv_{(1)} - u^{(1)}v_{(2)} + \int v_{(2)}\frac{du^{(1)}}{dx}dx$$

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